

Two spectral-resource regimes for autonomous temporal information

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Abstract

Temporal information in an autonomous quantum system need not be accompanied by global time-translation asymmetry: a clock and signal can carry a relative temporal mode while their joint state remains inside a fixed total-energy sector. What physical resource then limits the information in that mode? We first show that local Fisher information alone has no universal high-frequency energy bound: at fixed mean energy it can remain finite while the affine neighborhood over which its tangent is physical collapses. This exposes two complementary regimes. At nonzero physical tangent radius, arbitrary finite-copy collective measurements obey a two-sided spectral-survival bound. At a rank-changing boundary, the resource changes order: positive second-order endpoint synthesis is required, yielding sharp autonomous action bounds with coefficients $\hbar\nu/4$ for bilateral exchange and $\hbar\nu/2$ for one-sided synthesis. The laws extend to coherent baselines and multiple exchange frequencies, and fixed-total-energy constructions saturate both the single-gap constants and the complete frequency-weighted sum. Thus relational temporal information remains resource constrained even when global asymmetry and signed energy curvature vanish.

I. INTRODUCTION

Temporal information is usually discussed relative to an external time standard: a laboratory clock defines when a signal arrives, how quickly a probe evolves, or which phase a waveform carries. For a fully autonomous quantum description that reference cannot be supplied for free. A finite clock, signal, controller, and readout must instead be treated as parts of one physical system. This raises a basic resource question: *what must a finite autonomous system possess in order to carry information about relative time at a given frequency?*

The question is not answered by global time-translation asymmetry alone. In Page–Wootters-type relational descriptions, nontrivial relative evolution can occur inside a state that is stationary under the total Hamiltonian [1]. Resource theories of asymmetry organize the harmonic coherence required to break time-translation symmetry [2], and shared asymmetry has been used to quantify resources in relational-time constructions [3]. Quantum Fisher information provides the standard local statistical geometry for quantum estimation [4] and also has an operational resource interpretation for energetic coherence [5, 6], while quantitative Wigner–Araki–Yanase and conservation-law results constrain operations that

require a coherence reference [7, 8]. These frameworks are essential context, but they leave open a different local question. A clock and signal may exchange an exact energy gap while the total state remains stationary, so the global asymmetry can vanish identically even though a relative temporal mode is statistically resolvable.

A second obstruction appears if one tries to bound that mode directly by Fisher information and mean energy. We construct a two-level family whose mean excess energy is fixed while the Fisher information of an arbitrarily high Bohr-frequency tangent remains constant. What collapses is instead the radius of the affine parameter neighborhood over which that tangent remains a positive density operator. Thus a purely local metric does not retain enough state-space geometry to support an energy–frequency law. This observation leads to the first resource regime: when the affine physical tangent radius is nonzero, the robust combination of Fisher information and radius is bounded by the population of the spectral endpoints that support the transition. In an autonomous clock–signal exchange the same quantity is bounded on both local sides, even for a globally stationary baseline with pre-existing relational coherence.

The affine radius can, however, vanish at the boundary of state space while an exact nonlinear family remains physical. A first-order amplitude can enter a previously empty spectral sector provided the population needed for positivity is created at second order. Rank-changing quantum Fisher and Bures geometry is known to exhibit precisely such singular behavior [9]. Here we ask for its operational temporal-resource consequence. We show that the missing resource moves from zeroth-order spectral survival to positive second-order spectral synthesis. For one endpoint, arbitrary finite-copy collective measurements obey a Fisher–curvature bound. When opposite endpoint orientations participate in the same temporal mode, their score amplitudes combine by a sharp Minkowski law rather than by ordinary addition. In an exact autonomous exchange this produces a two-sided synthesis-action bound that remains nonzero inside a fixed-total-energy eigenspace, where both global time-translation asymmetry and signed total-energy curvature vanish.

The resulting picture is a two-regime resource principle for relative temporal information. At nonzero affine radius, information is backed by *pre-existing spectral survival*; at a rank-changing zero-radius boundary, it is backed by *spectral synthesis action*. Both statements hold for arbitrary finite copy number and arbitrary collective POVMs. Their coefficients are sharp in finite-dimensional fixed-total-energy models. We further show that the boundary

law survives arbitrary coherent support through a support/endpoint geometric reduction and that multiple exchange frequencies share one positive Hessian budget; a single Fourier measurement simultaneously saturates the complete frequency-weighted sum in a mode-separated fixed-energy shell.

These results are complementary to, rather than replacements for, established asymmetry and metrology bounds. We do not propose a new generic resource theory of time, and the synthesis action is not identified with the total dynamical energy required to implement an encoding. Protocol-level energy-constrained metrology addresses that broader implementation cost [10], while waveform Holevo theory treats attainable information under general linear waveform estimation [11]. Our claim is narrower: for a finite autonomous system carrying an exact relative temporal mode, there are frequency-resolved necessary state-family resources that remain operative when global asymmetry is zero, and the relevant resource changes order at a rank-changing boundary.

II. PROBLEM, CONVENTIONS, AND CLAIM BOUNDARY

We work throughout with finite-dimensional clock and signal Hilbert spaces. This is the finite-internal-system setting of interest here and makes the support pseudoinverses, compressed endpoint eigenvalues, and shorted-subspace constants used below unambiguous. Infinite-dimensional extensions require additional domain, closed-range, and spectral-regularity hypotheses and are not claimed.

We consider local two-parameter quantum statistical models whose two real coordinates are the physically fixed cosine and sine quadratures of one temporal mode. Let A denote the corresponding complex tangent and define

$$D_c = \frac{A + A^\dagger}{2}, \quad D_s = \frac{A - A^\dagger}{2i}. \quad (1)$$

For a POVM $\{M_y\}$ at the baseline state ρ_0 , write

$$p_y = \text{Tr}(\rho_0 M_y), \quad z_y = \text{Tr}(A M_y). \quad (2)$$

The outcome derivatives are $\partial_c p_y = \text{Re } z_y$ and $\partial_s p_y = \text{Im } z_y$, so our two-quadrature classical Fisher trace is

$$\text{Tr } F_1 = \sum_{y: p_y > 0} \frac{|z_y|^2}{p_y}. \quad (3)$$

For N independently encoded copies, F_N denotes the Fisher matrix of an arbitrary joint POVM; no separability or asymptotic assumption is imposed unless stated explicitly. For multiple modes, the coordinate space is the direct sum of these cosine–sine planes with its Euclidean metric. Accordingly, the Fisher and Hessian traces below refer to this fixed physical quadrature normalization; invariance under arbitrary anisotropic reparameterizations is not claimed.

An operator A_ν is an exact positive Bohr-gap tangent of frequency $\nu > 0$ for Hamiltonian H when

$$[H, A_\nu] = \hbar\nu A_\nu. \quad (4)$$

For an autonomous clock–signal pair the corresponding exact exchange condition is

$$[H_S, A_\nu] = +\hbar\nu A_\nu, \quad [H_C, A_\nu] = -\hbar\nu A_\nu, \quad (5)$$

which implies $[H_C + H_S, A_\nu] = 0$.

The resource statements below are deliberately distinct from general resource theories of time-translation asymmetry, the Page–Wootters mechanism, quantitative Wigner–Araki–Yanase constraints, waveform Holevo bounds, and protocol-level energy-constrained quantum metrology. Those frameworks supply essential prior structure [1–3, 5, 6, 8, 10, 11]. Our target is narrower: frequency-resolved necessary resources for a local *relational* temporal Fisher mode, including rank-changing physical families for which global time-translation asymmetry can vanish identically.

Definition 1 (Linear physical tangent radius). *For a baseline ρ_0 and complex tangent A , $R_{\text{lin}}(\rho_0, A)$ is the supremum of $R \geq 0$ such that*

$$\rho_0 + xD_c + yD_s \succeq 0 \quad \text{for every } x^2 + y^2 \leq R^2. \quad (6)$$

If no nonzero disk is physical, $R_{\text{lin}} = 0$.

The affine radius is not the radius of an arbitrary nonlinear physical family. A rank-deficient model can have $R_{\text{lin}} = 0$ while a nonlinear state family remains physical because second-order population is generated automatically.

III. FIXED MEAN ENERGY DOES NOT BOUND UNRESTRICTED LOCAL FISHER INFORMATION

Proposition 1 (Fixed-energy high-frequency local-Fisher counterexample). *Fix an excess mean energy $E > 0$ and a constant $c > 0$. For every $\nu > E/\hbar$, let*

$$H_\nu = \hbar\nu|1\rangle\langle 1|, \quad \rho_0 = (1-p)|0\rangle\langle 0| + p|1\rangle\langle 1|, \quad p = \frac{E}{\hbar\nu}, \quad (7)$$

and choose $A_\nu = 2c|1\rangle\langle 0|$. Then $\text{Tr}(\rho_0 H_\nu) = E$ for all ν , while an equatorial covariant qubit POVM has

$$\text{Tr } F_1 = 4c^2 \quad (8)$$

independently of ν . The affine physical radius is

$$R_{\text{lin}}^2 = \frac{p(1-p)}{c^2} \xrightarrow{\nu \rightarrow \infty} 0. \quad (9)$$

Hence no baseline-energy-only law can force arbitrary local Fisher information to vanish at high frequency without an additional physical-robustness or finite-amplitude assumption.

The counterexample identifies the variable lost by a purely local metric: the state-space neighborhood over which the tangent itself remains physical.

IV. FINITE-RADIUS SPECTRAL-SURVIVAL LAWS

Assume first that ρ_0 is stationary under H and A_ν obeys Eq. (4). Let $P_{U,\nu}$ and $P_{D,\nu}$ be the upper- and lower-endpoint projectors supporting the range and domain of the $+\nu$ mode, and define

$$U_\nu = \text{Tr}(\rho_0 P_{U,\nu}), \quad D_\nu = \text{Tr}(\rho_0 P_{D,\nu}). \quad (10)$$

Theorem 1 (Robust Fisher-survival law). *If $R_{\text{lin}} > 0$, then for every finite N and every joint POVM on N independently encoded copies,*

$$\boxed{\frac{R_{\text{lin}}^2}{4} \frac{\text{Tr } F_N^{(\nu)}}{N} \leq \min\{D_\nu, U_\nu\}.} \quad (11)$$

If H is semibounded with lower edge E_ and*

$$T(\nu) = \Pr_{\rho_0}[H - E_* \geq \hbar\nu], \quad (12)$$

then

$$\frac{R_{\text{lin}}^2}{4} \frac{\text{Tr } F_N^{(\nu)}}{N} \leq T(\nu), \quad (13)$$

and for finite mean excess energy $\bar{E}^+ = \text{Tr}[\rho_0(H - E_*)]$,

$$\bar{E}^+ \geq \frac{\hbar\nu R_{\text{lin}}^2}{4} \frac{\text{Tr } F_N^{(\nu)}}{N}. \quad (14)$$

Proof core. Positive affine radius forces $A = PAP$ on $P = \text{supp } \rho_0$. With $B = \rho_0^{-1/2} A \rho_0^{-1/2}$, positivity of the entire affine disk is equivalent to $R_{\text{lin}} w(B) \leq 1$, hence $\|B\| \leq 2/R_{\text{lin}}$ by the standard numerical-radius inequality [12]. Outcome-wise weighted Hilbert–Schmidt Cauchy–Schwarz gives $\text{Tr } F_1 \leq \text{Tr}(A \rho_0^+ A^\dagger)$. The latter operator is bounded by $4\rho_0/R_{\text{lin}}^2$ and supported on the upper endpoint. Applying the same argument to $A^\dagger$ gives the lower endpoint. For N copies, all off-diagonal copy terms in the weighted norm vanish because $\text{Tr } A = 0$, so the bound scales exactly as N even for an arbitrary collective POVM. The full derivation is given in the Supplemental Material.

A. Globally stationary relative time

Let C and S have semibounded local Hamiltonians and suppose

$$[\rho_0, H_C + H_S] = 0, \quad (15)$$

while A_ν obeys Eq. (5). Separate stationarity under H_C and H_S is not required for the support-sensitive extension. Define

$$T_X(\nu) = \Pr_{\rho_0}[H_X - E_{X,*} \geq \hbar\nu], \quad X \in \{C, S\}. \quad (16)$$

Theorem 2 (Autonomous dual-survival law). *For every finite N and arbitrary joint POVM,*

$$\boxed{\frac{R_{\text{lin}}^2}{4} \frac{\text{Tr } F_N^{(\nu)}}{N} \leq \min\{T_C(\nu), T_S(\nu)\}.} \quad (17)$$

Therefore

$$\bar{E}_C^+ + \bar{E}_S^+ \geq \frac{\hbar\nu R_{\text{lin}}^2}{2} \frac{\text{Tr } F_N^{(\nu)}}{N}. \quad (18)$$

Proof core. The signal Hamiltonian sees A_ν as a positive $+\nu$ gap; the clock Hamiltonian sees $A_\nu^\dagger$ as a positive $+\nu$ gap. Conjugation reverses only the sine coordinate, so both views have the same affine radius and Fisher trace. Applying the arbitrary-baseline upper-tail

form of the preceding proof on the two local sides gives Eq. (17). Thus a globally stationary relative mode is backed by matched local spectral survival rather than global time-translation asymmetry.

V. RANK-CHANGING BOUNDARY: SECOND-ORDER SYNTHESIS REPLACES SURVIVAL

Let $\rho(x, y)$ be a two-sided C^2 physical family with $\rho(0, 0) = \rho_0$, let $P = \text{supp } \rho_0$ and $Q = \mathbb{I} - P$, and let its first derivatives be Eq. (1).

Theorem 3 (One-sided boundary synthesis). *Suppose $P_U P = 0$ and $A = P_U A P$. Define*

$$J(A|\rho_0) = \text{Tr}(A\rho_0^+ A^\dagger), \quad T_U(x, y) = \text{Tr}[P_U \rho(x, y)]. \quad (19)$$

Then for every finite N and arbitrary collective POVM,

$$\boxed{\frac{\text{Tr } F_N}{N} \leq J(A|\rho_0) \leq \Delta T_U(0)}, \quad (20)$$

where $\Delta = \partial_x^2 + \partial_y^2$.

Proof core. The same weighted score inequality gives $\text{Tr } F_N/N \leq J$. For a one-parameter two-sided C^2 physical curve, Schur-complement positivity at a rank-deficient baseline implies $Q\rho''(0)Q \succeq 2K\rho_0^+ K^\dagger$, where $K = Q\rho'(0)P$. Applying this independently to the cosine and sine directions, whose support-to-kernel blocks are $A/2$ and $A/(2i)$, gives $\Delta T_U(0) \geq J$. This PSD-cone geometry is established mathematics [13, 14]; the resource consequence is the frequency-resolved positive endpoint synthesis it enforces.

For a pure boundary tangent with $PAP = QAQ = 0$, write

$$X = QAP, \quad Y = QA^\dagger P, \quad A = X + Y^\dagger, \quad (21)$$

and

$$J_+ = \text{Tr}(X\rho_0^+ X^\dagger), \quad J_- = \text{Tr}(Y\rho_0^+ Y^\dagger). \quad (22)$$

Theorem 4 (Bilateral boundary Minkowski law). *For every finite N and arbitrary collective POVM,*

$$\boxed{\sqrt{\frac{\text{Tr } F_N}{N}} \leq \sqrt{J_+} + \sqrt{J_-}}. \quad (23)$$

If X and Y enter mutually orthogonal baseline-empty endpoint sectors P_+ and P_- , with $T_{\pm}(x, y) = \text{Tr}[P_{\pm}\rho(x, y)]$, then

$$\boxed{\frac{\text{Tr } F_N}{N} \leq \left[\sqrt{\Delta T_+(0)} + \sqrt{\Delta T_-(0)} \right]^2.} \quad (24)$$

The square-root composition is sharp; ordinary additive endpoint-curvature accounting is false in general.

Proof core. For each POVM outcome the complex score is the sum of a right-supported X score and the conjugate right-supported Y score. Their weighted ℓ_2 norms are bounded by $\sqrt{J_+}$ and $\sqrt{J_-}$, so Hilbert-space Minkowski gives Eq. (23). In the PSD curvature inequalities the cosine/sine support-to-kernel blocks are $(X + Y)/2$ and $(X - Y)/(2i)$; adding the two operator inequalities cancels their cross terms exactly and yields a single shared curvature satisfying $C_{\Delta} \succeq Z_+ + Z_-$. Projection onto orthogonal endpoint sectors gives Eq. (24). An exact three-level Fourier model saturates the square-root law and violates naive additive curvature by a factor of two.

VI. AUTONOMOUS ZERO-RADIUS SYNTHESIS ACTION

Return to Eq. (5) and assume $PAP = 0$. In a clean spectral geometry, X synthesizes a signal-upper and clock-lower endpoint, while Y synthesizes a signal-lower and clock-upper endpoint. Define

$$\mathcal{A}_S^{(2)} = \frac{\hbar\nu}{4}(\Delta T_{S,+} + \Delta T_{S,-}), \quad \mathcal{A}_C^{(2)} = \frac{\hbar\nu}{4}(\Delta T_{C,+} + \Delta T_{C,-}). \quad (25)$$

These are positive absolute-gap endpoint-incidence resources, not signed subsystem-energy curvatures.

Theorem 5 (Autonomous dual synthesis-action law). *For every finite N and arbitrary collective POVM, bilateral synthesis obeys*

$$\boxed{\mathcal{A}_C^{(2)} + \mathcal{A}_S^{(2)} \geq \frac{\hbar\nu}{4} \frac{\text{Tr } F_N}{N},} \quad (26)$$

while one-sided synthesis obeys

$$\boxed{\mathcal{A}_C^{(2)} + \mathcal{A}_S^{(2)} \geq \frac{\hbar\nu}{2} \frac{\text{Tr } F_N}{N}.} \quad (27)$$

Both coefficients are sharp already at one copy.

Proof core. In the bilateral case, Eq. (24) and $(\sqrt{a} + \sqrt{b})^2 \leq 2(a + b)$ imply separately $\mathcal{A}_X^{(2)} \geq (\hbar\nu/8)(\text{Tr } F_N/N)$ for $X = C, S$. Adding the two local inequalities gives Eq. (26). If only one support orientation is present, the one-sided boundary theorem applies directly on each local side and doubles the coefficient. Thus the factor-of-two distinction is the operational consequence of whether two opposite boundary score amplitudes can interfere constructively.

A. Fixed-total-energy sharpness

Let

$$|L\rangle = |2_C, 0_S\rangle, \quad |M\rangle = |1_C, 1_S\rangle, \quad |U\rangle = |0_C, 2_S\rangle, \quad (28)$$

with equal local level spacing $\hbar\nu$, and take

$$\rho_0 = |M\rangle\langle M|, \quad A_\nu = c(|U\rangle\langle M| + |M\rangle\langle L|). \quad (29)$$

The exactly normalized family

$$|\psi(x, y)\rangle = \sqrt{1 - \frac{c^2}{2}(x^2 + y^2)} |M\rangle + \frac{c}{2}(x + iy)|L\rangle + \frac{c}{2}(x - iy)|U\rangle \quad (30)$$

is physical on $x^2 + y^2 < 2/c^2$ and remains inside the total-energy- $2\hbar\nu$ eigenspace. Each local endpoint Laplacian is c^2 , a three-outcome Fourier measurement gives $\text{Tr } F_1 = 4c^2$, and Eq. (26) is saturated exactly. The one-sided coefficient is saturated by the convention-consistent fixed-shell family

$$|\psi(x, y)\rangle = \sqrt{1 - c^2(x^2 + y^2)} |D\rangle + c(x - iy)|U\rangle. \quad (31)$$

VII. ARBITRARY COHERENT SUPPORT: MIXED SURVIVAL AND SYNTHESIS

Let $P = \text{supp } \rho_0$, $Q = \mathbb{I} - P$ and suppose A_ν obeys Eq. (5). First-order physicality gives $QA_\nu Q = 0$. Define

$$B = PA_\nu P, \quad K_+ = QA_\nu P, \quad K_- = QA_\nu^\dagger P, \quad (32)$$

and

$$Z_B^+ = B\rho_0^+ B^\dagger, \quad Z_B^- = B^\dagger \rho_0^+ B, \quad (33)$$

$$Z_+ = K_+ \rho_0^+ K_+^\dagger, \quad Z_- = K_- \rho_0^+ K_-^\dagger. \quad (34)$$

Write $J_B^\pm = \text{Tr } Z_B^\pm$ and $J_\pm = \text{Tr } Z_\pm$.

The support-preserving term need not itself remain an exact Bohr-gap operator when P fails to commute with the local Hamiltonians. Its resource therefore contains the geometry between its information-bearing support and the compressed physical endpoints. Shorting the local endpoint compressions onto the ranges of $Z_B^\pm$ gives candidate scalar survival ceilings. A local-side ceiling is available only when the corresponding shorting constant is strictly positive; a zero shorting constant gives no finite scalar ceiling from that side. Let a_+ and a_- denote the minima over the available finite upper- and lower-oriented local-side ceilings. If $B = 0$, set $a_+ = a_- = 0$. If no finite ceiling exists for a nonzero internal orientation, that orientation of the scalar mixed theorem is unavailable. Explicit formulas are given in the Supplemental Material. Shorted-operator mathematics is standard [15].

For the boundary contribution define canonical endpoint-role projectors of the full exact exchange,

$$\Pi_{\text{out}} = \text{supp}(A_\nu A_\nu^\dagger), \quad \Pi_{\text{in}} = \text{supp}(A_\nu^\dagger A_\nu), \quad (35)$$

and

$$G_{\text{ex}} = 2\hbar\nu Q(\Pi_{\text{out}} + \Pi_{\text{in}})Q. \quad (36)$$

For a C^2 family define

$$C_\Delta = Q(\partial_x^2 + \partial_y^2)\rho(0)Q, \quad \mathcal{A}_{\text{ex}}^{(2)} = \frac{1}{4} \text{Tr}(G_{\text{ex}}C_\Delta), \quad e = 4\mathcal{A}_{\text{ex}}^{(2)}. \quad (37)$$

For each nonzero synthesized orientation let $R_\pm = \text{supp } Z_\pm$ and

$$g_\pm = \lambda_{\min}\left(R_\pm G_{\text{ex}} R_\pm \Big|_{R_\pm}\right). \quad (38)$$

If a nonzero synthesized orientation has $g_\pm = 0$, the chosen scalar action does not control that direction: no finite action-only bound on its $J_\pm$ follows from $\mathcal{A}_{\text{ex}}^{(2)}$ alone.

For $a, e \geq 0$ and $p, q > 0$, define the bilateral scalar envelope

$$\Psi_a(e; p, q) = \begin{cases} (\sqrt{a} + \sqrt{e/q})^2, & e \leq ap^2/q, \\ (e + pa)(p^{-1} + q^{-1}), & e \geq ap^2/q. \end{cases} \quad (39)$$

Theorem 6 (Noncommuting autonomous mixed resource law). *Suppose first that both synthesized orientations are nonzero, $g_+, g_- > 0$, and both internal ceilings a_+, a_- are finite. Then for every finite N and arbitrary collective POVM,*

$$\boxed{\frac{\text{Tr } F_N}{N} \leq \min\{\Psi_{a_+}(e; g_+, g_-), \Psi_{a_-}(e; g_-, g_+)\}}. \quad (40)$$

If only the + synthesized orientation is present and $g_+ > 0$,

$$\boxed{\frac{\text{Tr } F_N}{N} \leq \min \left\{ a_+ + \frac{e}{g_+}, \left(\sqrt{a_-} + \sqrt{\frac{e}{g_+}} \right)^2 \right\}}, \quad (41)$$

where only branches with finite $a_{\pm}$ are retained. If only the - orientation is present and $g_- > 0$,

$$\boxed{\frac{\text{Tr } F_N}{N} \leq \min \left\{ \left(\sqrt{a_+} + \sqrt{\frac{e}{g_-}} \right)^2, a_- + \frac{e}{g_-} \right\}}, \quad (42)$$

again retaining only available finite branches. With no synthesized component,

$$\boxed{\frac{\text{Tr } F_N}{N} \leq \min\{a_+, a_-\}}, \quad (43)$$

where the minimum is taken over available finite internal ceilings.

Proof core. Weighted score Minkowski gives

$$\sqrt{\frac{\text{Tr } F_N}{N}} \leq \min \left\{ \sqrt{J_B^+ + J_+} + \sqrt{J_-}, \sqrt{J_B^- + J_-} + \sqrt{J_+} \right\}. \quad (44)$$

Shorting bounds each available $J_B^{\pm}$ by $a_{\pm}$. The canonical positive operator G_{ex} prices the same kernel curvature only once and yields $g_+ J_+ + g_- J_- \leq e$ with absent orientations omitted. Maximizing each bilateral Minkowski branch over this two-variable linear budget gives Eq. (39). Setting one $J_{\pm}$ to zero gives Eqs. (41) and (42) directly, without assigning a restricted eigenvalue to an absent range. In the clean pure-boundary one-sided limit, $a_+ = a_- = 0$ and $g = 2\hbar\nu$, so the specialized formula recovers $\mathcal{A}_{\text{ex}}^{(2)} \geq (\hbar\nu/2)(\text{Tr } F_N/N)$. The bilateral clean limit gives the $\hbar\nu/4$ coefficient.

The operator G_{ex} counts endpoint *roles*. If one energy state lies in both Π_{in} and Π_{out} , it is counted twice because it plays two roles. Equation (37) is a kinematic kernel-resolved endpoint-incidence action, not generally a signed mean-energy curvature or total protocol energy. In the shared-kernel qutrit benchmark the resource reduction gives exactly 12, while the SLD-QFI trace and exact common-record Fisher supremum are 43/4 and 55/8, respectively; the latter is proved by an explicit quadratic witness in the Supplemental Material.

VIII. MULTI-GAP SPECTRAL-ACTION SUM

Consider one common C^2 multiparameter family with zero-radius pure-boundary exact exchange modes A_k at positive gaps ν_k . The coordinates (x_k, y_k) are the fixed cosine/sine

quadratures introduced above. Define

$$X_k = QA_kP, \quad Y_k = QA_k^\dagger P, \quad (45)$$

$$Z_{k,+} = X_k \rho_0^+ X_k^\dagger, \quad Z_{k,-} = Y_k \rho_0^+ Y_k^\dagger, \quad (46)$$

and

$$C_\Sigma = Q \sum_k (\partial_{x_k}^2 + \partial_{y_k}^2) \rho(0) Q. \quad (47)$$

Second-order positivity gives the shared operator law

$$\boxed{C_\Sigma \succeq \sum_k (Z_{k,+} + Z_{k,-}).} \quad (48)$$

Let $G \succeq 0$ be one positive kernel cost and define

$$\mathcal{A}_{G,\Sigma}^{(2)} = \frac{1}{4} \text{Tr}(GC_\Sigma). \quad (49)$$

For every nonzero orientation let $g_{k,\pm}$ be the minimum restriction of G to $\text{supp } Z_{k,\pm}$. Define the effective mode price

$$\gamma_k = \begin{cases} (g_{k,+}^{-1} + g_{k,-}^{-1})^{-1}, & Z_{k,+}, Z_{k,-} \neq 0 \text{ and } g_{k,+}, g_{k,-} > 0, \\ g_{k,+}, & Z_{k,+} \neq 0, Z_{k,-} = 0, g_{k,+} > 0, \\ g_{k,-}, & Z_{k,-} \neq 0, Z_{k,+} = 0, g_{k,-} > 0, \\ 0, & \text{some nonzero orientation has zero restricted cost.} \end{cases} \quad (50)$$

A zero tangent mode may simply be omitted.

Theorem 7 (Multi-gap boundary action sum). *For one fixed arbitrary joint POVM on N independently encoded copies, let $F_{N,k}$ be the 2×2 Fisher block for (x_k, y_k) . Then*

$$\boxed{\sum_k \gamma_k \frac{\text{Tr } F_{N,k}}{N} \leq 4\mathcal{A}_{G,\Sigma}^{(2)}} \quad (51)$$

If a nonzero information-bearing orientation has zero restricted cost, its $\gamma_k = 0$ contribution is deliberately vacuous: a different G or the operator-valued curvature is needed for positive control of that mode. In a clean bilateral autonomous geometry where G assigns cost $2\hbar\nu_k$ to each orientation,

$$\boxed{\mathcal{A}_{G,\Sigma}^{(2)} \geq \sum_k \frac{\hbar\nu_k}{4} \frac{\text{Tr } F_{N,k}}{N}.} \quad (52)$$

Proof core. Apply the two real-coordinate PSD-cone inequalities to each mode and add the cosine and sine inequalities before taking any scalar resource. Cross terms cancel within each mode, and summing gives Eq. (48). Pricing once with G gives a common linear budget for all $J_{k,\pm}$. Modewise Minkowski followed by weighted Cauchy–Schwarz converts each positively priced bilateral pair to its harmonic price γ_k ; one-sided modes use their single price, while a cost-null information direction has zero scalar price. Summing yields Eq. (51). Thus the resource is one shared Hessian, not a sum of independently assigned curvature budgets.

A. Simultaneous sharpness with one common record

Take the fixed-total-excitation shell

$$|n\rangle \equiv |m-n\rangle_C |m+n\rangle_S, \quad n = -m, \dots, m, \quad (53)$$

with baseline $|0\rangle$ and

$$A_k = c_k(|k\rangle\langle 0| + |0\rangle\langle -k|), \quad \nu_k = k\omega_0. \quad (54)$$

The exact common family is

$$\begin{aligned} |\psi(\mathbf{x}, \mathbf{y})\rangle = & \sqrt{1 - \frac{1}{2} \sum_{k=1}^m c_k^2 (x_k^2 + y_k^2)} |0\rangle \\ & + \sum_{k=1}^m \frac{c_k}{2} (x_k + iy_k) |-k\rangle + \sum_{k=1}^m \frac{c_k}{2} (x_k - iy_k) |k\rangle. \end{aligned} \quad (55)$$

For $d = 2m + 1$, the discrete Fourier basis

$$|v_j\rangle = \frac{1}{\sqrt{d}} \sum_{n=-m}^m e^{in\phi_j} |n\rangle, \quad \phi_j = \frac{2\pi j}{d}, \quad (56)$$

gives

$$F_{x_k x_l} = 2c_k^2 \delta_{kl}, \quad F_{y_k y_l} = 2c_k^2 \delta_{kl}, \quad F_{x_k y_l} = 0. \quad (57)$$

Thus $\text{Tr } F_{1,k} = 4c_k^2$ for every mode simultaneously. With

$$G = 2\hbar\omega_0 \sum_{n \neq 0} |n| |n\rangle\langle n|, \quad (58)$$

Eq. (52) is saturated term by term and in the complete weighted sum. The Supplemental Material gives the discrete-harmonic orthogonality calculation explicitly.

For overlapping mode ranges in a general model, Eq. (51) remains valid for any supplied positive G , but a unique frequency-diagonal physical interpretation is not claimed.

IX. RELATION TO PRIOR WORK AND SCOPE

The broad mathematical and conceptual ingredients are established. Page and Wootters showed how relational evolution can be represented within a stationary closed-system description [1]. Resource theories of asymmetry characterize harmonic modes and coherence needed to break a symmetry [2]; shared asymmetry has been applied directly to the Page–Wootters mechanism [3]. Quantum Fisher information has an operational role as a resource of energetic coherence [5, 6], and conservation-law constraints on measurements and operations are treated by quantitative WAY and coherence-cost results [7, 8]. Rank-changing QFI/Bures singularities are known [9], while multiparameter and waveform-estimation theory separates SLD information from common-record attainability [11, 16, 17]. Energy-constrained metrology can charge the total energy consumed by preparation, controls, and measurement [10].

The present resource is different in scope. Equations (17) and (26) concern a frequency-resolved relational mode that can be globally time-translation symmetric. The finite-radius law charges pre-existing local spectral survival. At a rank-changing boundary, an entropy-type shared-asymmetry resource is generally nonanalytic in amplitude, whereas the positive endpoint-incidence action supplies a finite quadratic necessary coefficient. The action is kinematic: it is computed from the local Hessian of the encoded state family. We do not claim that it equals the total physical energy required to synthesize that family dynamically.

X. DISCUSSION

The theorem chain separates two mechanisms that a local Fisher metric alone conflates. In the interior of state space, a temporal mode can be strong only over a finite affine neighborhood; multiplying Fisher information by that physical radius exposes a pre-existing spectral-survival cost. At a rank-deficient boundary, the affine neighborhood can collapse even though a nonlinear family remains physical. The missing population must then appear at second order, and its positive spectral incidence supplies the boundary resource. In an autonomous exchange both mechanisms are two-sided because every signal energy step is matched by an opposite clock step.

The result is not a universal implementation-energy theorem and not a replacement for

asymmetry resource theory. It is a local, frequency-resolved necessary-resource theorem for finite-dimensional encoded relative temporal information. Important open problems remain: deriving a dynamical implementation cost that produces the kinematic action, extending the sharp spectral sum to controlled continua, and determining whether covariance-changing Gaussian families require an additional resource beyond endpoint curvature.

AI USE DISCLOSURE

OpenAI ChatGPT was used substantively as a research and manuscript-preparation tool. Manuscript-stage analysis used the GPT-5.6 Sol model; earlier project sessions used other GPT-5.6-series configurations, accessed in August 2026. The AI assisted with exploratory derivations, counterexample searches, literature synthesis, symbolic and algebraic checking, generation and debugging of numerical validation code, adversarial review, and drafting and revision of manuscript text. The author directed the work through explicit problem formulations and iterative review decisions. AI outputs were treated as provisional: candidate results were subjected to explicit analytic proofs, separate re-derivation passes, targeted counterexample searches, source verification, and numerical validators where applicable. The author retains full responsibility for the scientific claims, citations, and final manuscript.

DATA AVAILABILITY

This is a theoretical study; no experimental or observational data were created or analyzed. The manuscript source files, theorem-development notes, and custom numerical validation scripts supporting the reported results are publicly available in Ref. [18]. The numerical scripts provide consistency checks for the analytic statements; the proofs in the article and Supplemental Material are the primary basis for the claims.

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